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CIS 521

Project implementation

NEKHLAWI

نخلاوي

Midterm Report

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1. REPORT INTRODUCTION

The fast evolution of technology and artificial intelligence has also offered new possibilities to enhance the classic industry like agriculture. In Saudi Arabia and the Gulf area, date palm trees are a very significant agricultural and cultural asset that has a share in the food security and the economy of the region. Nevertheless, palm tree diseases are a significant problem to most of the farmers particularly the small scale farmers who might not have access to professional advice or state of the art diagnostic equipment. The diseases can be detected at early stages and treated, which will help a great deal to reduce the loss of crops and enhance the health of palm trees.

The current report details the development of the Nekhlawi application, which is a smart mobile application to assist palm tree farmers by offering AI-based disease detection and agricultural advice. Applying the image processing and machine learning methods, the application performs the analysis of palm tree pictures and determines the potential diseases. According to the analysis findings, the system offers treatment suggestions and information that can assist farmers to manage their crops in a better manner.

Besides the AI diagnosis, Nekhlawi will also establish contact with farmers and the agricultural experts via consultation services. The farmers would be able to seek professional assistance, schedule appointments, and have access to a source of knowledge where they would find valuable information about diseases and remedy approaches of palm trees, as well as the optimal farming techniques. The system also has a history of diagnoses and consultation to allow users to trace their activities and decision over time.

This report explains the changes and developments that have taken place since the proposal phase of the project. It also points out the modifications on the system design, the progress of implementation, the challenges encountered during the development process, and the testing plan that is set to be done on the system. The knowledge and skills gained in the process of developing the project are also reflected in the report.

2. SUMMARY OF PROPOSAL

During the proposal phase, the Nekhlwai application went through planning, analysis, and preliminary design of the interfaces. Proposing an application powered by AI that aims to detect defects in date palms using image processing and machine learning combined, and connect farmers with experts and knowledge. The application provides treatment recommendations and a history of both AI and expert consultations, reducing cost and increasing efficiency. The knowledge hub aims to increase awareness and protect rich traditions.

A discussion of different approaches and AI models was presented during the proposal phase, using a comprehensive review of related literature along with evaluation. The review focused on date palms and detecting their diseases, to align with the 2030 vision in Saudi Arabia in agricultural sustainability and technology transformation. Different strategies for handling data, including data collection, preprocessing, and feature extraction, were also discussed. By the end of the literature review, we conducted a gap analysis to guide future improvements.

Furthermore, the proposal defines the project's overall objectives, motivation, scope, limitations and implications related to implementing the Nekhlwai application, such as legal, ethical, and social implications, etc. Moreover, it presents the software project management plan (SPMP), which highlights roles and responsibilities, resources and budget, scheduling, and risk management. Moving on to the Software Requirements Specification (SRS) and Software Design Specification (SDS), where we specified system requirements, both functional and non-functional, how the application is structured, database attributes and design, use cases, and UI design. However, nekhlawai is expected to be a solution that aims to provide faster disease detection and treatment, improve accuracy and reduce losses, and enable communication between experts and farmers to support the smart agriculture field in the Gulf region.

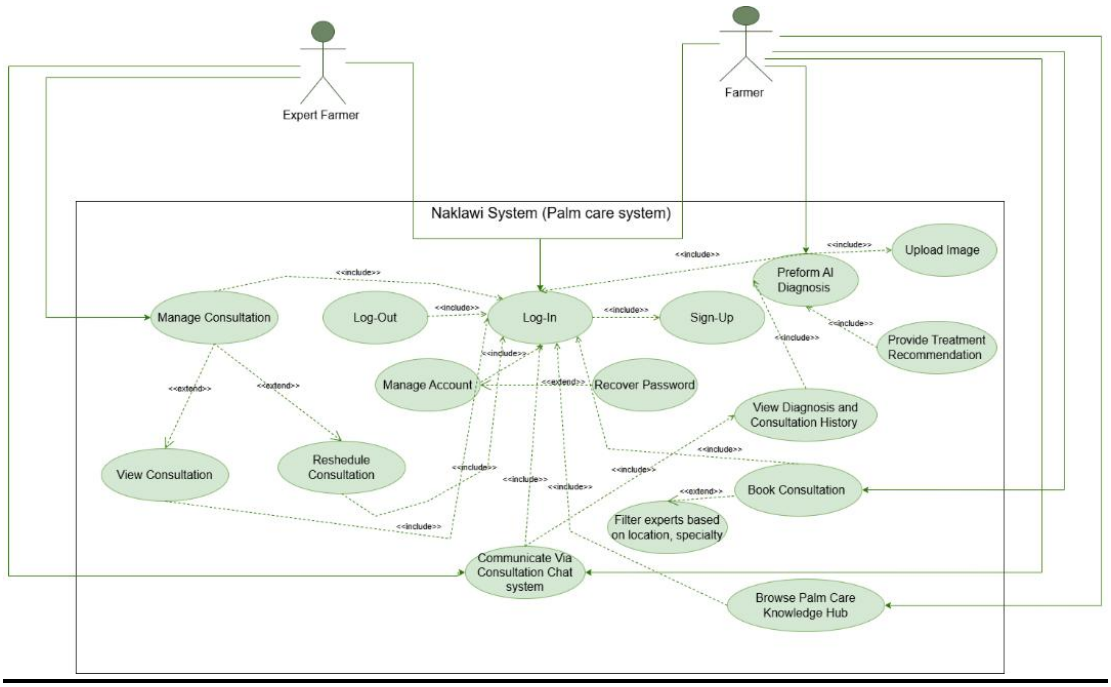
3. CHANGES FROM PROPOSAL

1. Features Flow Update

- **Reason:** We enhanced the flow of some features to make it more usable to the users.

2. Use-Case Diagram Modification

- **Reason:** The "Include" relationship was incorrect. Each use case requires a login use case.



3. Database Design Update

- **Reason:** The DB design was enhanced to improve system organization.

4. Class Diagram Modification

- **Reason:** The class diagram was updated to reflect recent DB and system flow changes (features, relationships and data structures).

5. Missing Interfaces Modification

- **Reason:** New interfaces were added and some existing interfaces were enhanced to improve user usability and experience. Including Book Consultation and its related pages (Book Consultation, Select Timeslot, Past & Incoming Consultations, Rate Expert).

6. **Development Tools Update**
   - **Reason:** Supabase and Android Studio development tools were adopted to improve backend integration and development efficiency.

7. **Supported Platforms Update**
   - **Reason:** The application scope was expanded to support both Android and iOS devices instead of being limited to iOS only, increasing accessibility for users.

4. IMPLEMENTATION PROGRESS

The Nekhlawi project implementation is concerned with how the system design and requirements can be changed into a working mobile application. In this step, the team was involved in development of various elements of the system, such as the mobile application interface, backend services, and the artificial intelligence model that was applied to detect palm disease.

The mobile application started to be developed with Flutter and Dart programming language. A number of user interface screens have been introduced to enable users to interface with the system easily. Such screens consist of the welcome page, the pages of the login and registration, the page of the AI diagnosis, and the knowledge hub area. The design was based on simplicity and ease of use so that farmers can simply navigate and use the application.

Second, the team had been busy in the preparation of the backend infrastructure required to support the functionality of the system. Such services as Firebase and Supabase were researched to handle data storage, authentication, and communication between the parts of the application. The database design was also introduced depending on the design presented in the Software Design Specification (SDS) such as the tables containing users, consultations, messages, payments, and disease information.

The other significant aspect of the implementation procedure was the creation of the AI model that would detect palm tree diseases. The team examined various machine learning models and tested lightweight models like MobileNetV2 and EfficientNet-B0 to ascertain that the model can be effectively run in mobile devices. This model was trained on available datasets and deployed with the help of TensorFlow and Keras with the aim of converting it to TensorFlow Lite in order to implement it in a mobile environment.

Also, the team planned and applied a number of the missing user interface pages, which were absent in the former phase, including consultation booking page, time slot selection page, consultation history page, and expert rating page. The pages were created to finalize the consultation process and enhance the user experience.

All in all, the project is making progress towards developing a working prototype of the Nekhlawi application. Some parts of the application are yet to be developed and integrated, but the team has managed to get the key framework needed to support the application, which includes mobile interface, database design, and preparation of the AI model.

CHAPTER 5: SYSTEM IMPLEMENTATION

AI MODELS

Nekhlawi software incorporates one deep learning model to ensure reliable diagnosis of palm tree leaves and provide actionable steps to solve the problem.

To achieve the best performance and accurate score we conduct comparative training sessions between three different models.

1.1 Data Preprocessing

Before training, we have collected data from different resources [في ملف الداتا مذكورة هناك اللي استخدمته]. Which we end with 9150 then was preprocessed. This includes resizing images to 240X240, normalization, and data augmentation techniques (such as rotation and flipping) to improve accuracy percentage.

1.2 Model comparison

- EfficientNetB0 with 99.78% accurcy .
- MobilenetV2 with 98.29% accury.

1.3 Final Selection

After comprehensive training, EfficientNetB0 was selected as the primary model for Nekhlawi. This model was chosen for its superior accuracy.

SYSTEM PAGES AND FEATURES

2.1 Welcome Pages

Before using the application, the user is presented with a Welcome Page (Figure 1), followed by three consecutive pages that provide a brief introduction of the application's key features (Figure 2).

Figure 2: Welcom Page

Figure 1: App Introduction Page

2.3 Sign In (Farmer and Expert)

Both farmers and experts can log in through a shared page by entering their email and password or using the magic link.

Figure 3 : Sign In Page

2.2 Sign Up (Farmer and Expert)

Both farmers and experts can create a new account on the Nakhlawi app. The user begins by selecting the account type, as shown in Figure 3, and then enters their email address and password. After Supabase verifies that the email address is not already in use with the app, it sends a confirmation email (Figure 4). Clicking the link redirects the user to the app to complete the remaining information and agree to the terms and conditions (Figure 5).

Figure 4: Role Selection Page

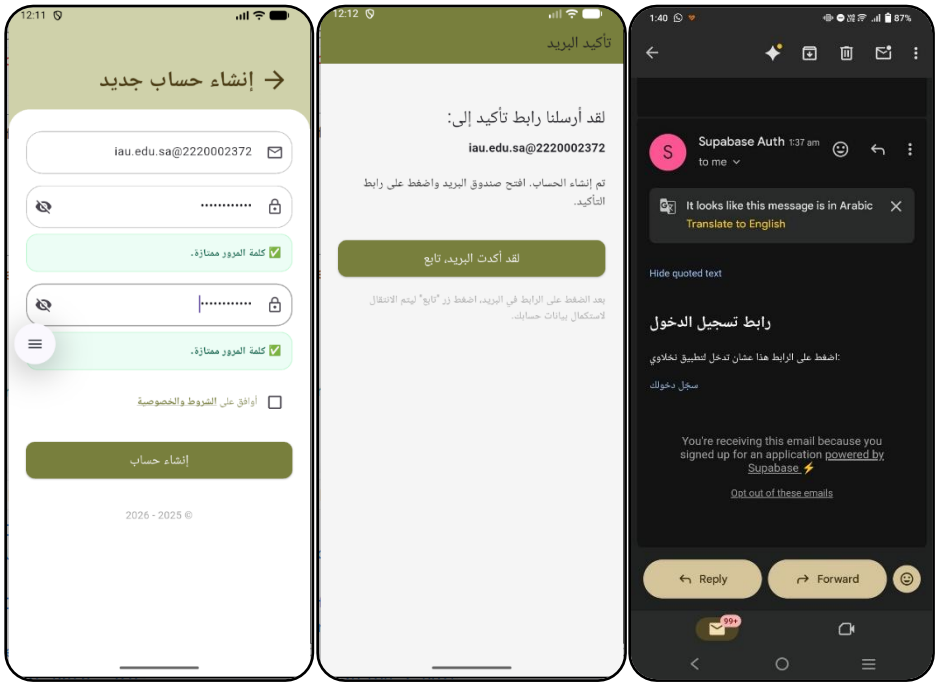


Figure 5 : Sign Up Pages

Figure 6 : Terms and Conditions Page

2.4 Forgot Password (Farmer and Expert)

If the farmer or expert forgets their password, they can reset it via a reset link sent to their original account email.

Figure 7: Forgot Password

2.5 Home Page

After logging in, the user is redirected to the homepage, which features a simple welcome message and a clear presentation of all the application's services. This facilitates user interaction for new users and reduces the learning curve. At the top of the page, all consultations booked by the user are displayed. If no consultations are booked, the application indicates this with a message, shown in Figure 8.

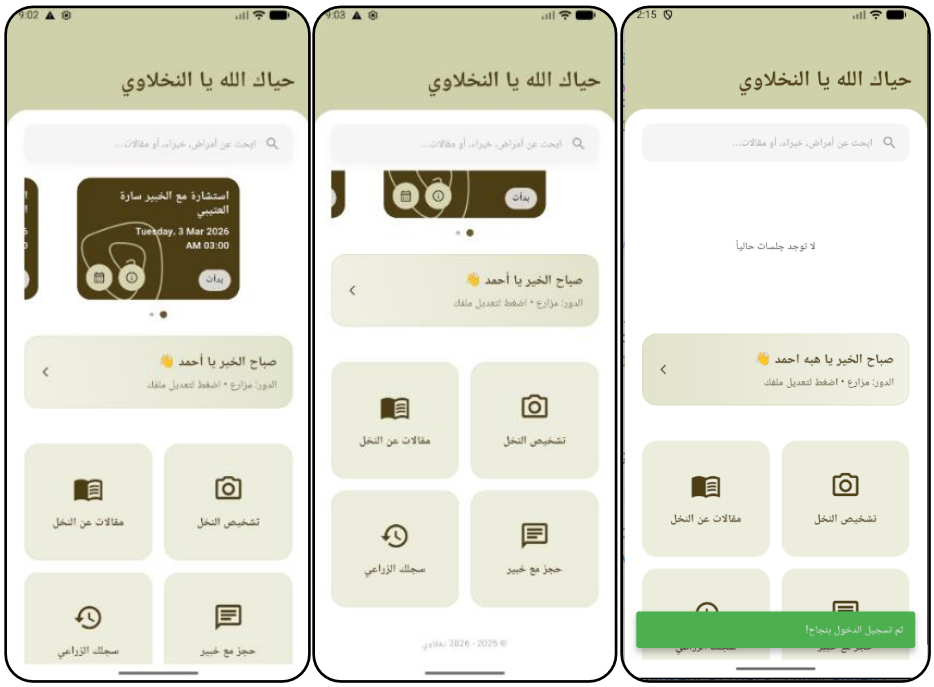


Figure 8 : Home Page

2.6 Account & Profile Management (Farmer)

The farmer can edit and manage account data on the profile page. Any new information added will be reflected directly in the database.

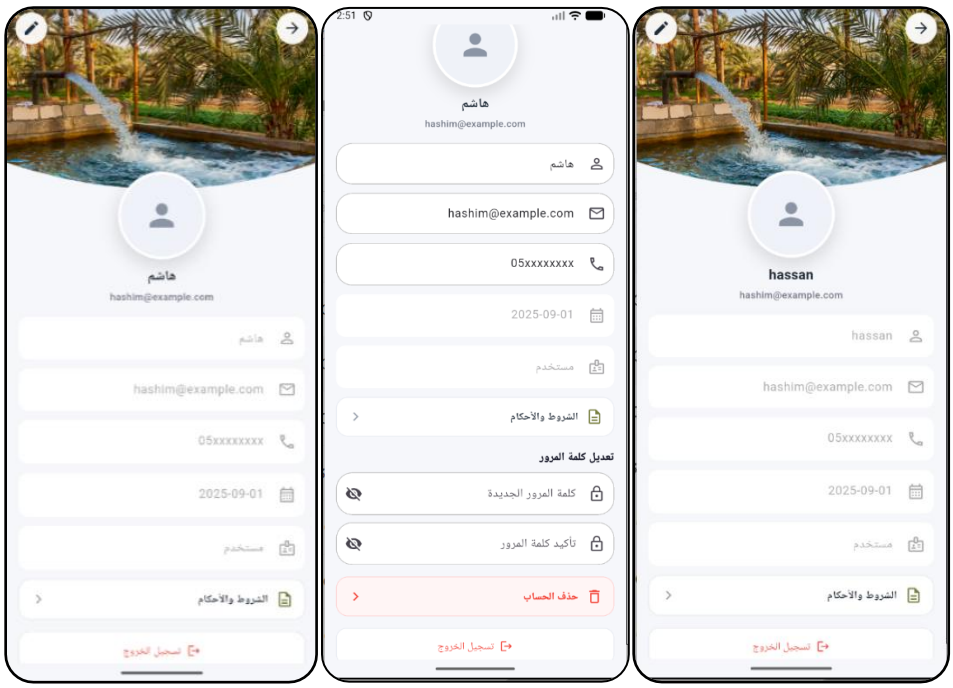


Figure 9: Account Page

2.7 Logout (Farmer)

On the same profile page, the user can log out of the account. The system will automatically redirect them to the login page to log back in or create a new account.

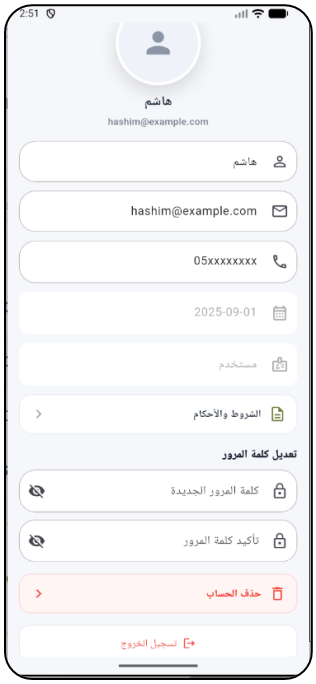


Figure 10 : Logout

2.8 Delete Account (Farmer)

On the same profile page, when editing the profile, the user will see the option to delete their account. If this option is selected, a pop-up will appear asking the user to confirm their desire to delete the account. If confirmed, the account will be permanently deleted, along with all data associated with it in the application's database.

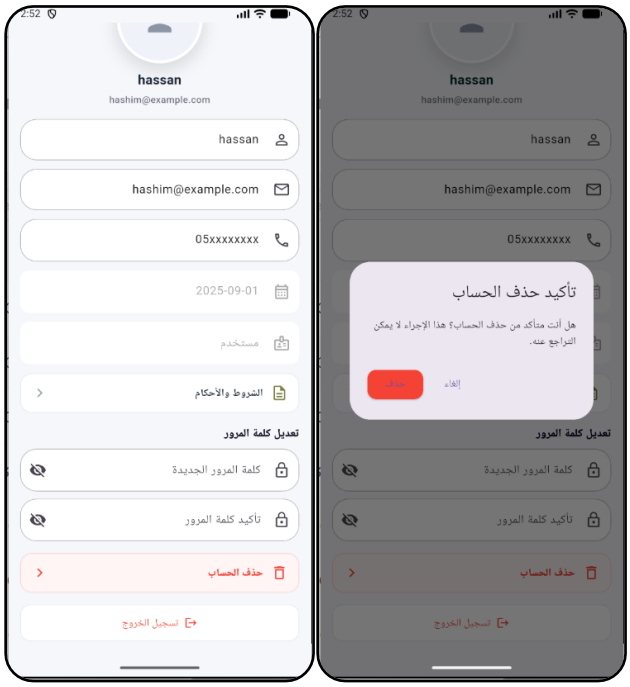


Figure 11 : Delete Account

2.9 AI-Based Diagnosis & Treatment Recommendation

If the user selects the palm tree diagnosis service, the system will request permission to access the phone's camera and photos in order to take or upload an image of the affected palm tree. Before starting the diagnosis, a pop-up window will appear with instructions on how to upload the image, after which the user can take or upload the photo.

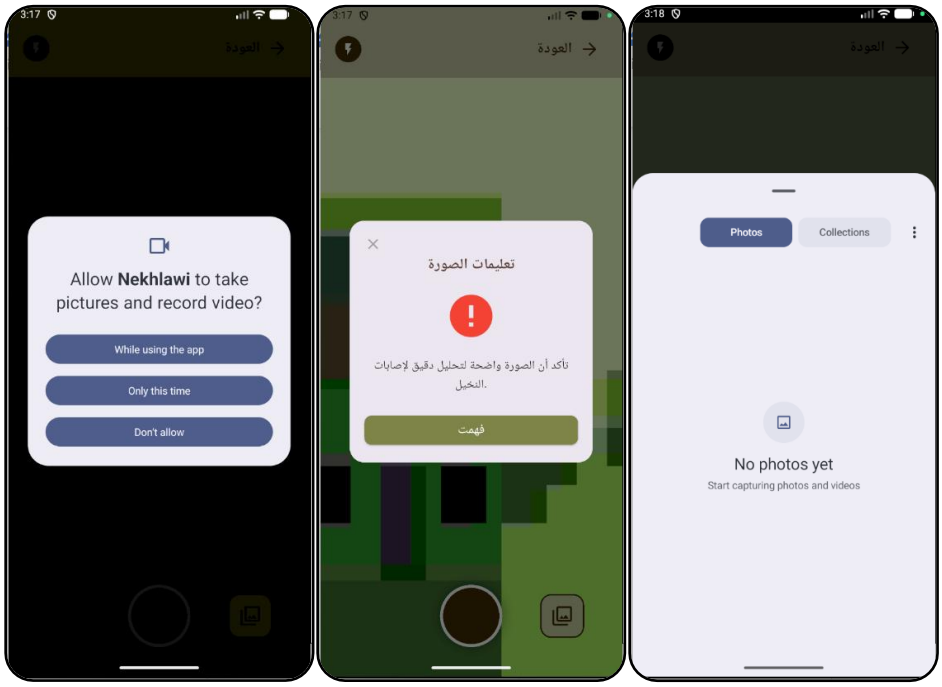


Figure 12: AI-Based Diagnosis & Treatment Recommendation

2.10 Consultation Booking (Farmer)

If a user chooses to book a consultation, a list of all experts registered in the system will appear, along with the option to view their profiles and see other users' comments about them. When a user selects to book with an expert, a list of all available booking times will appear.

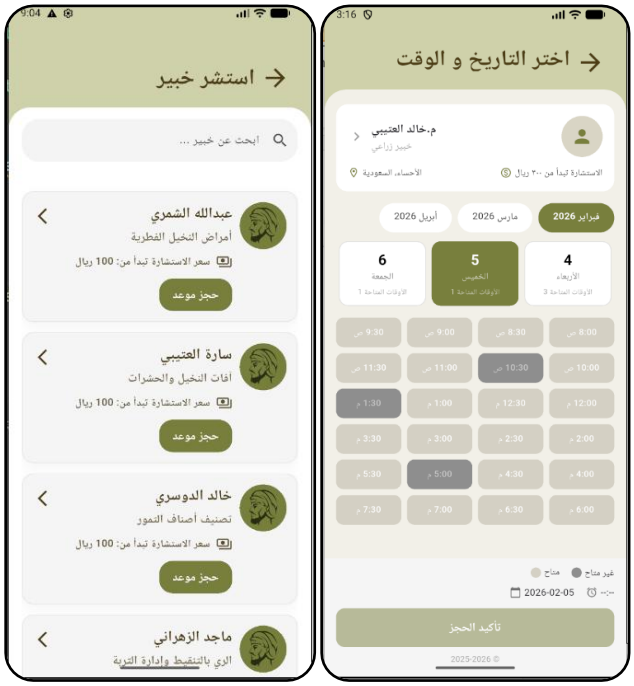


Figure 13: Consultation Booking

2.11 Consultation Management & Scheduling

This page is currently under development.

2.12 Diagnosis & Consultation History Management

After completing a consultation session with an expert or an AI consultation, the user can access these consultations through the history on the homepage and view the consultation results at any time.

Figure 14 : History Page

2.13 Palm Care Knowledge Hub (Mini Wiki)

When the user selects the mini-wiki, a page appears filled with short articles about the date palm and all related topics (such as dates, care, the importance of the palm, and more). Selecting any title takes the user to the article page with full details.

Figure 15 : Mini Wiki Page

2.14 Manage Reviews (Expert)

This page is currently under development.

CHAPTER 6: TEST PLAN

1. INTRODUCTION

Nekhlawi is an Android application designed to bridge the gap between traditional palm cultivation and care and technology in Saudi Arabia. This app testing plan involves testing and evaluating app features:

1. Account & Profile Management.
2. AI-Based Diagnosis & Treatment Recommendation.
3. Consultation Booking.
4. Consultation Management & Scheduling.
5. Diagnosis & Consultation History Management.
6. Palm Care Knowledge Hub.

1.1 Objectives

The main goal of this test plan is to track the testing processes within the planned schedule to ensure that the application works easily, with good quality and efficiency. All existing features will be tested by using a multi-level testing strategy (Unit, Integration, System) for waterfall SDLC model. The work activity includes Test Case Specification preparation, Test Environment Setup, Execution, Defect Reporting, and the composing of a final Test Summary Report. Budget Resources Needed for Nekhlawi will be determined by Project leader "Zainab Al-Badi". The major milestones will be associated with completion of features and the successful passing of entry/exit criteria for each level of testing. Total testing effort will consume around 30% of the total project development time which is reserved for validation.

1.2 Testing Strategy

The testing strategy of Nekhlawi will ensure the features of software item work well. A V-model approach will be used to ensure comprehensive software quality insurance through the system life cycle.

1. **Test levels:**
   - Unit testing: This phase will help to verify each individual components, methods, classes for Account & Profile Management.

- Integration testing: This phase expose faults in integrated components. At will use an incremental approach top-down methodology.
- System testing: This phase will evaluate the end-to-end system and its compliance with all specified functional and non-functional requirements.

2. **Test Techniques and Approach:**
   - White-Box testing: It's a type of Unit testing and Integration test to verify the internal structure and logic of the code. By using the line/statement to ensure every line of code will be executed at least once coverage and basis path test to design test cases to execute independent paths through the program's control flow graph, cyclomatic complexity test.
   - Black-Box testing: To verify functionality based on user requirements without looking at the internal code structure, that will be used during System testing, and UAT.
   - Automated testing: Will be integrated across the development lifecycle, focusing on efficiency and repeatable checks. By using Performance Testing and Regression Testing.
   - Testing Environments: Development Environment: Unit Integration and system testing will be performed within the development environment, using testing frameworks and mock data to isolate components for testing.

3. **Risks and Constrains:**
   - Technical Risks: Due to limited knowledge of integration and connecting AI model, project data, cloud-based services would increase the system failure
   - Data & Accuracy Risks: The chance of the AI model providing a false diagnosis because of the limited images or environmental noise.
   - Timeline Constrain: The project must be completed within specified timeframes.

1.3 Scope

The testing plan will examine all features that exist in the Nekhlawi Android application:

- The testing process will evaluate AI-based systems through their ability to identify pests and diseases in image inputs.
- The testing process will evaluate three functions of the consultation platform which include real-time chat and video calls and appointment scheduling with agricultural experts.
- The system will verify educational resource accessibility through its ability to present articles and infographics to users.
- The system will verify user account security through its ability to protect account creation and profile updates and history management processes.

The testing plan will not extend beyond its current boundaries which already restrict testing to palm crops and large-scale farm ERP systems.

1.4 Reference Material

- Software Test Plan (STP).
- Software Project Management Plan (SPMP).
- Software Requirements Specification (SRS).
- Software Design Specification (SDS).

1.5 Definitions and Acronyms

Table 1: Definitions and Acronyms

Acronyms	Abbreviation	Definition
STP	Software Test Plan.	It's a formal document used to describe the scope, approaches, activities for software testing.
SPMP	Software Project Management Plan.	It's a formal document used from the beginning and end for managing, controlling the software development.

Acronyms	Abbreviation	Definition
SRS	Software Requirements Specification.	It's a comprehensive document to outline the software purpose, functions, and constraints of software.
STR	Software Test Report.	Its formal report that summarizes the results of the entire testing efforts that have been done in a software project. It provides a detailed account on the coverage of the test, reduction of the success and failure of each test, and the overall quality of the software at a specified time.
SDS	Software Design Specification.	It's a comprehensive document that act as blueprint for system, where it translates the requirements into technical roadmap.
SDLC	Software Development Life Cycle.	A chart illustrates a systems development life cycle that is a summary of the key stages to be undertaken alongside the delivery products in the computerized system validation.
UAT	User Acceptance Testing.	An assessment process whereby the system conformance to established acceptance requirements is followed, allowing the customer to evaluate the compliance of the system to their acceptance levels.
AI	Artificial intelligence.	The computing technology used in the application that will identify pests and diseases by using images as inputs and produce the correct treatment as guidelines.

Acronyms	Abbreviation	Definition
API	Application Programming Interface.	This is a collection of back-end services and methods of exchange between the mobile application, and the cloud or database infrastructures.
CR	Change Request.	An officially written request that is given when an unexpected change in requirements or technical requirements is discovered.
RTM	Requirements Traceability Matrix.	This is a methodical device applied by the development team to assess the effects of the change and to confirm that all the requirements have been tracked to its respective test cases.
ERP	Enterprise Recourse Planning.	Wholesome business management applications: the project specification at Nekhlawi clearly states that the scope of testing will not include large-scale farm enterprise resource planning systems.
VS Code	Visual Studio Code.	Its advanced text editor complete development platform which is a cross-platform Integrated Development Environment (IDE). It serves as the main workspace where developers can write, debug, as well as perform version control of their code.

Acronyms	Abbreviation	Definition
Colab	Google Collaboratory.	It is a Jupyter Notebook on cloud-based environment. It supports literate programming through allowing executable code, rich textual annotations and data visualizations to co-exist together in a single document.
-	Flutter.	Google created the UI (User Interface) Software Development Kit which uses a single codebase to draw native-appearing and acting widgets on both iOS, Android, Web, and desktop platforms.
-	Dart.	It is enacted in a client-streamlined object-oriented programming language and was created to be easy to study besides offering high-performance when it comes to the undertakings that are bound to user interface dialogs.
-	Firebase.	A Business-as-a-Service model of Google. It has a set of cloud-based services including NoSQL databases and authentication systems, which means that developers do not have to maintain their own servers.
-	Super-base.	A BaaS alternative based on open source backends. Despite having similar features to Firebase, it is developed on a PostgreSQL, which is a traditional relational database system.

2. TEST ITEMS

This section in the test plan will cover all functional and non-functional components of Nekhlawi system and ensure that all modules comply with the project's Requirements Specification, Design Specification and User Guide.

2.1 Program Modules

In program modules, Nekhlawi Application includes 5 main modules:

1. **User Account & Authentication:** In this module we will test the system ability of verifying and handling valid and invalid inputs in registration, sending and confirming of OTP and successful creation of new user accounts, log in credentials, password recovery and secure access to the right type of account and profile information.

2. **AI Diagnosis Module:** In this module we will test the system capability to capturing/uploading palm images, processing the image using the AI model, showing diagnosis results with the confidence score, retrieving a treatment recommendations, and proper handling of low-confidence or failed diagnosis results.

3. **Consultation Module:** In this module we will test the system capability of showing available experts, searching experts by name or specialization, filtering consultation results, verifying time slot availability, accepting payments, updating consultation status, sending notifications, supporting session approval, cancellation, and rescheduling, and facilitating real-time communication between users and experts.

4. **Knowledge Hub (Mini Wiki) Module:** In this module we will test the system ability of loading and displaying educational content related to palm trees, searching articles using keywords, displaying images and videos correctly inside the content, navigation between articles, and handing cases when no search results are found.

5. **History Module:** In this module we will test the system capability of storing and accessing past AI diagnosis and consultation history, searching within history records using keywords, and presenting the items of history, opening the full details of a particular diagnosis or consultation.

2.2 Job Control Procedures

Not applicable to the Nekhlawi system.

2.3 User Procedures

Testing the User Guide includes:

1. Reviewing it to ensure it clear explanation of all system functions.
2. Ensures the including of all the screenshots that match the actual interface.
3. Verifying instructions are accurate, up-to-date, and easy to follow.
4. Checking that all steps are complete and no essential guidance is missing.

2.4 Operator Procedures

Not applicable to the Nekhlawi system.

3. FEATURES TO BE TESTED

Table 2: Features To Be Tested

Req ID	Feature	Applicable Role	Priority (1-5)	Description
User Account & Authentication Module				
1	Role Selection	Both (Expert user – Non expert user)	5	The user must choose the type of account (an expert user who provides consultations - an unexpert user who reserves the consultations) before creating an account and based on his selection, the fields are displayed to be filled out by the user
2	Sign-up	Both	5	After choosing a role, the user will fill in the fields and after that a new account will be created and saved in the data base
3	Log-in	Both	5	The user logs in to the application using the registered email and the password, a magic link is sent to the registered email, when the user clicks on it, the system redirects the user to the home page in the application and displays its data

Req ID	Feature	Applicable Role	Priority (1-5)	Description
User Account & Authentication Module				
4	Log-out	Both	5	The user can log out successfully from his accounts, the system redirects the user to the sign up page
5	Forget Password	Both	5	The user can change the password before logging in. The user enters the registered email, and the system will send a magic link to the email, when the user clicks on it, the system redirects the user to the page of changing the password inside the application and chang the password
6	Delete Profile	Both	5	The user can delete his account and all his data from the data base completely
7	Profile Management	Both	5	The user can update his profile information and change the account password
8	Error Messages Clarity	Both	4	If the user enters any data that system shall validate user input and display an appropriate error message if the entered data does not match the required format

Req ID	Feature	Applicable Role	Priority (1-5)	Description
User Account & Authentication Module				
9	Image Upload / Camera Capture	Both	4	The user can upload or capture clear palm images for diagnosis successfully
10	AI Disease Detection	Both	5	AI model analyzes the uploaded image and returns the predicted disease
AI Diagnosis Module				
11	Result Display Page	Both	4	The system displays the AI prediction to the user
Consultation Module				
12	Booking Consultation	Non expert user	4	Non experts can schedule consultation sessions
13	Chat System	Both	4	Experts and non experts in the start of the scheduled can communicate with experts via real-time chat
14	Notification System	Both	2	The user receives reminder notifications before the consultation sessions begin and for messages and booking updates
15	Consultation Management	Both	3	The user can modify the consultation appointment

Req ID	Feature	Applicable Role	Priority (1-5)	Description
Consultation Module				
16	View Expert Profile	Non expert user	3	The user can view all the available experts and view their profiles (expert name, specialization, years of experience, bio, rating and past review)
17	Submit Rating	Non expert user	3	After the end of the consultation session user rate the expert and the session and can optionally write a review
Knowledge Hub (Mini Wiki) Module				
18	Article Display	Both	4	The user can view all the articles in the DB
19	Search Articles	Both	1	The user can search in the articles by the article title
History Module				
19	View Consultation History	Both	3	The user can view his past consultations
20	View Diagnosis History	Both	3	The user can view his past AI diagnosis
21	Search Consultation History	Both	1	The user can search in his history for any past consultations

Req ID	Feature	Applicable Role	Priority (1-5)	Description
History Module				
22	Search Diagnosis History	Both	1	The user can search in his history for any past AI diagnosis

4. FEATURES NOT TO BE TESTED

Table 3 : Features Not to Be Tested

Feature	Justification
Unsupported Devices & Versions	Nekhlawi application does not support outdated Android versions and older devices (Android 12 and older); hence, the testing plan will not cover them.
Third-Party Platform Failure	Our application will only integrate a mock of the external services. The failures actually caused by external third-party services, such as cloud services, will not be covered by our testing plan.
Cyber Attacks Simulation	The Nekhlawi app development team consists of students with limited resources and time constraints to implement advanced security testing.

Feature	Justification
Extreme Performance & Stress Testing	Stress testing with a high load of simultaneous users is not needed, as the target users are expected to be limited.

5. APPROACH

The testing process of Nekhlawi Mobile Application will be through systematic and scheduled activities to verify both functional and non-functional software requirements which should enable the proper estimation for completing each activity using both black box and white box methodologies.

Main testing activities will include: Component Testing, Integration Testing, Interface Testing, Security Testing, Recovery Testing, Performance Testing, Regression Testing and Acceptance testing.

where each is defined with the objectives, inputs, methods, criteria, and expected outputs. The tests will be done manually and with the help of some testing tools. This will be achieved through the use of a Requirement Traceability Matrix (RTM) to ensure requirement traceability, where each requirement is accompanied by what test cases and test results. This type of testing is used to make sure that all the activities of the system are addressed, defects are found and detected with minimal effort and that Nekhlawi meets performance, security, and user acceptability requirements during the release.

5.1 Component Testing

Component testing (unit testing) aim is to verify that each distinct module and unit functions correctly, while also confirming the accuracy, completeness, and alignment of the units with the design specifications. The development team will carry out the component testing by building test cases for white box testing methods on each individual module source code. A method of basis path testing will be applied to guarantee that every independent branch at each decision point is executed by using tools such as JUnit.

- **AI Diagnosis Module**

Table 4: Ai Diagnosis Module Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
A 5MB clear palm leaf Image from user's gallery	AI diagnosis & confidence rate	-	-	The diagnosis will be saved in history

Table 5: Ai Diagnosis Module Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
A captured palm leaf image in darkness	Message: "Image unclear. Unable to determine diseases accurately." And offers retaking the image or booking consultation	-	-	The diagnosis won't be saved in history
A 23MB palm leaf image from user's gallery	Message: "Image size is too large. please upload an image under 10MB"	-	-	

- **Consultation Module: Booking**

Table 6: Consultation Module: Booking Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
User clicks at expert > Expert available > Slot available > Connected > Payment Success	Mark booking as pending, waiting expert approval	-	-	-

Table 7: Consultation Module: Booking Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Select an unavailable expert	Message: "No expert available at the moment"	-	-	-
Select an unavailable time slot	"Selected time is no longer available. Please pick another slot"	-	-	-
Lost Connection while clicking "Confirm booking"	Message:"Unable to complete booking. Check your connection and try again"	-	-	-
Paying with insufficient funds	Message: "Unable to complete booking. Due to the payment is not complete"	-	-	-

- **Consultation Module: Management**

Table 8: Consultation Module: Management Successful Test Case

	Successful Test Case				
	Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Expert	Click “Approve booking” on a pending request	User receives "Consultation successfully approved" alert.	-	-	Status: "Approved"
	Click “Decline” on a pending request	User receives reschedule/selection prompt.	-	-	Status: "Declined"
	No consults	Display: "You don't have any upcoming consultations at the moment."	-	-	-
Farmer	View notification > click “Cancel”	Display confirmation prompt	-	-	Status "Canceled"
	View notification > clicks “Cancel” > clicks “Back”	Process stays in "Incoming chat" state	-	-	No status change
	View notification > clicks “Reschedule”	Redirect to "Book Consultation” page	-	-	Status "Rescheduled"

- **Consultation Module: Chat System**

Table 9: Consultation Module: Chat System Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Wait for session time > type “hello”, > send	Typed message sent and reached the other party	-	-	-

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Click “+” to add file > permission to gallery granted > upload “image.jpg”	Image is successfully uploaded and displayed in the chat.	-	-	-
Click "End Session"> click “Confirm” > submit rating	Chat closes and review saved in the data base	-	-	Status: "Complete"

Table 10: Consultation Module: Chat System Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Tap "Chat Consultation" before scheduled time	Chat remains locked/inactive	-	-	-
Click “+” to add file > Deny access to gallery	Message: "Media access was denied. You cannot send photos..."	-	-	-
Click “+” to add file > permission to gallery granted > upload “mypalm.exe”	Message "Unsupported file format. Please select a valid file."	-	-	-
Click “End session” > click “Cancel”	Conformation prompt closes and the chat session stays active	-	-	-

- **Knowledge Hub (Mini Wiki) Module**

Table 11: Knowledge Hub (Mini Wiki) Module Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Click at Article #1	Article #1 Displayed	-	-	-

- **History Module**

Table 12: History Module Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
User selects “history” icon > picks past Ai diagnoses> clicks “Export”	Display past diagnoses details (image/result/date) and exports the file	-	-	-
User selects “history” icon > picks past consultations > clicks “Export”	Display past consultations details (expert notes/date/ status) and exports the file	-	-	-

Table 13: History Module Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
User selects "history" icon with no history	Display empty page with text "No diagnosis or consultations found yet, start with scanning your palm tree or book consultation"	-	-	-
User selects "history" icon > picks past Ai diagnoses or consultation	Displays past Ai diagnosis/consultations details	-	-	"only viewing"
Click item > click "Export">click "Cancel" on conformation	Export process is terminated and redirect to the selected history page.	-	-	-

5.2 Integration Testing

The primary objective of Integration Testing is to ensure that distinct software modules within Nekhlawi system which have previously undergone testing and operate correctly when interfaced with other subsystems. It is important to ensure that the interfaces of the modules are performing as intended and that the overall system functions in accordance with the design and operational specifications. The methodology adopted will be Black box testing using the incremental approach or the top-down methodology and will be applied using tools such as Postman and Firebase Console.

- **Role selection**

Table 14: Role selection Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Clicks "Expert/Farmer" role button	Displays Expert/Farmer account creation form according to role	-	-	-

- **Sign up**

Table 15: Sign Up Successful Test Case

	Successful Test Case				
	Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Expert	Name: Ahmed Sumpul Email:AhmedSumpul@gmai.com Phone Number:9667329123 Years of Experince: 10 Specialty: Pests Password: 99223338S$ Confirm password: 99223338S$ Terms and conditions: accepted Captcha: accepted Code: 9999	Display Dashboard page	-	-	New Account information stored in database

	Successful Test Case				
	Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Farmer	Name: Mohammed Ahmed Email: Moh67@gmail.com Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted Captcha: accepted Code: 9999	Display Dashboard	-	-	-

Table 16: Sign Up Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Name: Mohammed Ahmed Email: **Moh67@gm//ail.com** Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted Captcha: accepted Code: 9999	Redirect to error field with error indication	-	-	-

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Name: Mohammed Ahmed Email: Moh67@gmail.com Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted Captcha: accepted **Code: Invalid**	Message: "Invalid or expired code"			
Email: Moh67@gmail.com Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted **Captcha: Unaccepted** Code: Invalid	Message: "Failed Captcha, redo again"			

- **Log Out**

Table 17: Log Out Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Clicks "log out" button	The user will exit the account and	-	-	-

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
	redirected to "log in" screen			

- **Forgot Password**

Table 18: Forgot Password Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Email: AhmedSumpul@gmail.com Completed Capcha Code: 0233 New password: 99223338S$	Message: "Password successfully reset"	-	-	-

Table 19: Forgot Password Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Email: Ahmed??a@gmail.com	Message: "Email not found"	-	-	-
Email: AhmedSumpul@gmail.com Uncompleted Captcha Code: 0233 New password: 99223338S$	Message: "Failed Captcha, redo again"	-	-	-

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Email: AhmedSumpul@gmail.com Completed Captcha Wrong Code	Message: "Invalid or expired code"	-	-	-
Email: AhmedSumpul@gmail.com Completed Captcha Code: 0233 New password: u99	Display error notification and password rules	-	-	-

5.3 Conversion Testing

Not Applicable.

5.4 Job Stream Testing

Not Applicable.

5.5 Interface Testing

Interface testing will be conducted to ensure that the Nekhlawi backend interacts effectively and as intended with the external systems such as payment gateways and account authentication services. It will be executed by the testing and the development team after the integration process with the external systems where it will involve the execution of API calls, the simulation of responses and the verification of error handling through applying tools such as Espresso.

5.6 Security Testing

Security testing is performed to ensure that the Nekhlawi system is able to enforce proper access control and validate being free of any major security vulnerabilities. The test will be performed by the quality assurance and testing teams after integration by using password vulnerability tests and role checks using tools like Firebase Console and Postman.

5.7 Recovery Testing

Recovery test will be conducted to ensure that Nekhlawi system recovery and backup processes work as intended to when any failures occur. The test will be conducted by the Testing Team after the backup procedures are configured by simulating failures (server crash, network loss) and ensuring the proper system recovery by using tools like Firebase Console.

5.8 Performance Testing

Performance testing is a process that evaluates the system efficiency of the Nekhlawi system while operating under different levels of user interaction. This is aimed at measuring the response time of the system, the stability of the system, and whether it can handle simultaneous interactions among users without delays and malfunctions. This will determine whether the system is performing within the set performance standards and the test scenarios will be used to measure the time needed by the system to respond to various activities, including navigations, booking, and checkouts.

5.9 Regression Testing

Regression testing is conducted to verify that any changes or enhancements made to the Nekhlawi system will not adversely impact features that have been already tested / were validated to function correctly to preserve the system's stability after the introducing any new functionalities, in appearance of any bugs, or modifications to any segments of the code or database. The test will be conducted using tools such as GitHub, Junit and Espresso.

5.10 Acceptance Testing

Acceptance testing is conducted as a component of quality assurance to verify that Nekhlawi satisfies all of its specified requirements, ensures that it operates as intended from the perspective of the end-user, and identifies defects and issues while it's still possible. Prior to deployment, the Product Owner will implement specified acceptance test cases aligned with SRS requirements, where End users engage in realistic scenarios (such as entering mini-wiki, reviewing history, utilizing AI diagnosis, and scheduling a consultation). These tests will transpire within a controlled User Acceptance Testing (UAT) environment that emulates the production setting and will culminate in the confirmation from end users. All acceptance criteria established within the SRS must be fulfilled without any outstanding critical defects.

5.11 Beta Testing

Not Applicable.

6. PASS / FAIL CRITERIA

The specified criteria for accepting or rejecting test case results are listed below:

- Pass: The result of test case execution matches the expected result, and the function works as specified in the system's requirements. Data is stored properly in the database and protected from unauthorized access.
- Fail: The test case execution result does not match the expected output; data is stored incorrectly, the function does not perform properly, or any other system deviation from the intended output occurs.

6.1 Suspension Criteria

Testing shall be suspended under the following conditions:

- Missing Components: Major parts of the system have not been built yet, leading to inaccurate testing results or making testing infeasible.
- Critical Defects: Defects that prevent the execution of the test process sequence, such

as a login failure.

- Backend Services Inaccessibility: Database connection is lost, or core API services do not respond.
- AI Model Accuracy: The confidence score of the AI model does not meet the required threshold point.

6.2 Resumption Criteria

Testing shall resume when the following conditions are met:

- sBuilt Components: Functions and code blocks necessary for testing have been built.
- Defect Resolution: The Quality Assurance team has verified that blocking defects are resolved.
- Backend Services Accessibility: All needed services are responding to the system's requests.
- AI Model Accuracy: The confidence score of the AI model meets the threshold point.

6.3 Approval Criteria

The following conditions must be met to approve the test results:

- Requirement Fulfillment: All system features mentioned in the SRS work as intended.
- Success Rate: At least 90% of the critical test cases succeed.
- Defect Density: Less than 10% of non-major defects are detected.
- Performance: The system meets benchmarking performance standards.
- Formal Sign-off: Formal approval from the stakeholder is received.

7. TESTING PROCESS

This section discusses the expected documentation, tasks to be completed, resources, responsibilities, and schedule to be followed.

7.1 Test Deliverables

By the end of the testing activities, the two documents below should be delivered:

• Software Test Plan (STP).

• Software Test Report (STR).

7.2 Testing Tasks

- Develop all required documents during the testing process
- Prepare the testing environment and assign resources and tools
- Define testing tasks, responsibilities and schedules
- Educate on testing cases and activities using the proper approaches
- Document and resolve defects
- Assure the quality of the software and check for compliance with requirements
- Manage any changes that occur

7.3 Responsibilities

Using the guidance of the team leader, all team members will be responsible for planning and conducting testing activities, coordinating resources, and following the schedule. Moreover, team members will be responsible for developing and assuring the quality of the software by resolving defects and verifying code quality. On the other hand, the academic supervisor is responsible for supervising and approving overall processes.

7.4 Resources

Table 20: Resources

Requirements	Resources
Human	• Team members • Supervisor

Hardware	• Personal laptops • Cloud-based processing
Software	• VS Code • Google collab • APIs • Libraries • Android studio • Firebase • Super-base • Google Drive • Flutter • Dart

7.5 Schedule

Table 21: Schedule

Testing Task	Start	End
Plan Development	08/02/2026	18/02/2026
Test Case Preparation	12/04/2026	16/04/2026
Setting Test Environment Components	19/04/2026	21/04/2026
Integration Testing	22/04/2026	24/04/2026
System Testing	25/04/2026	30/04/2026
Regression Testing	01/05/2026	03/05/2026

Testing Task	Start	End
Performance Testing	04/05/2026	06/05/2026
Security Testing	07/05/2026	08/05/2026
Recovery Testing	08/05/2026	09/05/2026
Acceptance Testing	10/05/2026	12/05/2026
Report Completion	13/05/2026	16/05/2026

8. ENVIRONMENTAL REQUIREMENTS

8.1 Hardware

The following hardware resources are required:

- Android smartphones (13 and above versions)
- high-performance computers
- Stable internet access
- Backup storage device for Nakhlawi application data
- Cloud server infrastructure

8.2 Software

The following software resources are required:

- Android Studio

- Firebase / Cloud backend services
- AI model framework (Tensorflow)
- Version control system (GitHub)
- Testing frameworks (JUnit, Espresso)
- Browser (Google Chrome / Edge) for backend dashboard testing.
- Database management system (Firebase Firestore or MySQL)

8.3 Security

The following security measures are required during testing:

- Backend systems access control
- Secure database transmission (HTTPS/SSL)
- Restricted access to AI model configuration files
- Test database backup and recovery mechanisms
- Secrecy of test user data to avoid unauthorized modification

8.4 Tools

The following tools will be used:

- **JUnit**

  It is used to perform unit testing of the Java components and check the logic of the program to verify that the various modules are working properly.

- **Espresso**

  It is applied to test the interfaces of Android applications automatically to test the

interactions between users.

- **Postman**

  Applied in API testing and verification of the backend services to ensure that the requests and responses are handled appropriately.

- **Firebase Console**

  To trace database activities, authentication, and backend log testing.

- **GitHub**

  Used to track changes, version control, and collaborative development management.

- **Performance Monitoring Tools**

  It is used to test response time, load performance of a system, and general application performance.

- **Microsoft Word**

  Test plans, test case specifications, defect reports, and final test summary reports are prepared and documented using it.

Training and Support:

The setup and efficient use of all testing tools will be assisted through official documentation, online tutorials, and user manuals by team members.

8.5 Publications

- Software Design Specification (SDS)
- Software Requirements Specification (SRS)
- API Documentation

- AI Model Documentation
- Android Development Documentation
- User Guide / User Manual

8.6 Risks and Assumptions

Table 22: Risks and Assumptions Table

#	Risks	Assumptions	Contingency Plans
1	**Low AI Accuracy** The AI model can be used to give wrong or low-confidence diagnoses because of limited training data or low-quality images	System testing is preceded by training the AI model on adequate and appropriate palm disease datasets	Expand the testing data size and range, change the level of confidence, and retrain or optimize the model (where needed)
2	**Time Constraints** The defect rework or integration delay can cause the testing activities to exceed the planned schedule	The development modules will all be accomplished on schedule	Select the important and risky test cases, use automated regression testing, and change the testing schedule where necessary
3	**Limited Testing Devices** Lack of Android devices can limit compatibility tests with various	The testing phase will be able to access required hardware resources	Test Android Emulator with various settings and get other devices where possible

	settings		
4	**Backend Service Downtime** APIs or cloud-based services can go unavailable at certain points of testing	Backend infrastructure and APIs will not be changed during testing and will be stable and accessible	Keep reserve test data and set up offline testing procedures that do not need to be connected to the cloud

9. CHANGE MANAGEMENT PROCEDURES

9.1 Change initiation

When we identify a chance of unplanned modification of Nekhlawi main requirements, or technical, we will prepare the change request document (CR) and then send it to Mrs. Albandari for approval throw the progress report in week seven.

9.2 Change review

Once the CR is complete, the team will analyze the effect of the change by using the Requirements Traceability Matrix (RTM), identify any resources needed, and be on track with the planned schedule.

9.3 Change authorization process

The project team will begin the Technical Realignment phase after we receive formal approval of the Change Request (CR). The development branch will undergo changes through the addition of new features and the removal of existing ones. The team of Quality Assurance will conduct Test

Suite Synchronization while developing new test cases for new features and deactivate test cases which test obsolete functions.

10. PLAN APPROVALS

This plan is approved by:

- Project Leader: Zainab Hassan Al-Badi
  Date: 1 / 3 / 2026

  Signature:

- Project Supervisor: Mrs. Albandary Mohammad Alamer
  Date: 1 / 3 / 2026

CHAPTER 7: DISSEMINATION PLAN

The paper is finalized and planned for submission to a relevant journal. Dissemination will follow upon publication.

LESSONS LEARNED

In Nekhlawi project implementation, the variety of modules and infrastructure components raised the need to learn multiple frameworks, programming languages, tools, and skills. We started with learning how to build the mobile app front end using Flutter and Dart by building the mobile app pages, designing them in high readability and usability to enhance user experience while simultaneously learning about the backend infrastructure, researching what components according to our need and budget and how to pick the right platforms and tools like using firebase and superbase. We also tried to learn more about machine learning frameworks like PyTorch and TensorFlow using Python programming language and researched what type of models suites our system more, we tried to train multiple models where we found that we needed light models like MobileNetV2 and EfficientNet-B0 with enough accuracy to be properly integrated with our mobile app system with proper results to meet our goal of efficiency. After making some research and advice seeking we decided to stick to using TensorFlow with Keras library to be able to use TensorFlow Lite and manage to integrate the chosen model properly with the app in flutter, this allowed us to explore more in the machine learning platforms and gave us a glance at what the machine learning field and mobile app development filed looks like and what does it offer. We also learned to use some collaborative tools like Google collab, GitHub.

ISSUES FACED

During the implementation of the project, we encountered several challenges. From a technical perspective, working with entirely new technologies with limited resources hindered the project's progress. Flutter was hard to install, and working on an incompatible operating system and emulator led to different issues throughout the implementation. Picking up the AI model, training it and linking it to the application without proper hardware devices was time-consuming and inefficient. From the managerial perspective, the leader puts great effort into dividing the work among team members and checking on the progress of tasks, yet the misalignment between team members schedule cause a communication delay and misunderstanding of tasks logical execution leading to redundant work and waste of efforts. However, the team addressed these gaps fast enough to improve efficiency and performance, minimize redundant work, and save efforts.

CONCLUSION

Finally, this report has outlined the current state of development of the Nekhlawi application since the proposal stage. The intended project will offer an intelligent palm tree disease detection and management solution through a combination of artificial intelligence, mobile software, and professional advice services.

During the implementation process, the team was engaged in building the application interface, the preparation of the backend infrastructure, and testing machine learning models that could be integrated with the mobile field. It also made some design enhancements, such as updating the database structure, use-case diagram, and adding a few user interface pages about consultation services.

Although the development is fraught with all the mentioned difficulties, including technical constraints, getting acquainted with new technologies, and team members coordination, the team was capable of overcoming the obstacles and proceed with its work on the system completion.

The Nekhlawi application can assist farmers by helping them detect diseases early, offering dependable advice on how to treat them, and having access to expert advice directly. The project helps to improve smart agriculture by introducing modern technologies and applying them in agriculture and corresponds to the purposes of technological development and sustainability in Saudi Arabia and the Gulf countries.

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